

Problem Statement

3+ Weeks

Average time MSEs spend
identifying applicable standards

564+

BIS standards in Building
Materials alone to navigate

₹Crores

Compliance penalty risk for
non-certified products

The Challenge

Indian Micro & Small Enterprises (MSEs) lack in-house expertise to navigate India's sprawling BIS standards ecosystem. Manual search is slow, error-prone, and costly — the wrong standard = product recall or legal liability.

Solution Overview

What We Built

An end-to-end RAG pipeline: type a product description → get the top-5 relevant BIS standards with rationale, in under 2 seconds.

Hybrid Retrieval

BM25 sparse + FAISS dense retrieval fused with Reciprocal Rank Fusion

Cross-Encoder Reranking

ms-marco MiniLM re-ranks top-10 candidates for precision

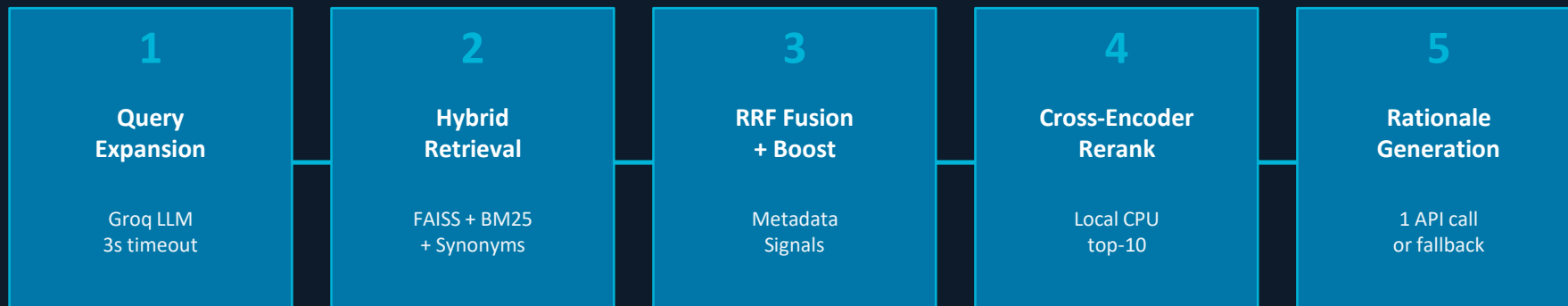
LLM Rationale

Single batched Groq/Llama call explains why each standard matches

Zero-API Fallback

Works fully offline — chunk-text rationale when no API key is set

System Architecture



Where each step runs:

Steps 2-4

100% LOCAL — no network, no API, no rate limits

Step 1 & 5

Optional Groq API — skipped gracefully if unavailable

inference.py

ALL LOCAL — LLM completely disabled for judge evaluation

Chunking & Retrieval Strategy

Chunking

- Source: BIS SP 21:2005 PDF — Building Materials summaries
- Strategy: One chunk per IS standard (semantic boundary)
- 564 chunks total — each 100–300 words
- Metadata enriched: category, grade, cement type, part number
- Why? Preserves IS number ↔ content association for accurate retrieval

Retrieval

- Dense** FAISS IndexFlatIP + all-MiniLM-L6-v2 (cosine similarity)
- Sparse** BM25Okapi with synonym expansion (OPC, PPC, AAC...)
- Fusion** Reciprocal Rank Fusion k=60 combines both rank lists
- Boost** Metadata signals add +0.15 per flag match (max +0.30)
- Rerank** Cross-encoder ms-marco-MiniLM scores top-10 pairs

Demo Highlights

Streamlit UI

"We manufacture 33 Grade OPC cement..."

#1 IS 269 : 1989

CEMENT

#2 IS 8042 : 1989

CEMENT

#3 IS 8112 : 1989

CEMENT

Key Features Shown in Demo

⚡ Sub-2s response from query to 5 results

🔍 Category tags (CEMENT, CONCRETE, AGGREGATE) on each result

📄 Expandable BIS chunk text with source context

🏆 Rank badges showing retrieval confidence order

💬 LLM-generated rationale: 'Why this standard?'

🔍 Graceful no-API fallback with chunk-derived explanation

Evaluation Results

100%

Hit Rate @3

Target: >80%

1.0

MRR @5

Target: >0.7

1.52s

Avg Latency

Target: <5s

Test Set: 10 queries from public test set | All targets exceeded on first run

Before vs After Optimization

Metric	v1 (naive pipeline)	v2 (this project)
Hit Rate @3	100%	100% ✓
MRR @5	1.0	1.0 ✓
Avg Latency	42.08s ✗	1.52s ✓
Retrieval	Dense only	Hybrid BM25 + FAISS + RRF
BIS x Sigma Squad AI Hackathon API calls/query	7 (rate limits)	0 in inference.py

Impact on MSEs

BEFORE

- ✗ 3+ weeks of manual BIS catalogue search
- ✗ Compliance consultants cost ₹5,000–₹20,000
- ✗ Wrong standard = product recall risk
- ✗ MSEs often skip certification entirely



AFTER

- ✓ Top-5 standards in under 2 seconds
- ✓ Zero cost — free to use, open source
- ✓ Rationale explains why each standard applies
- ✓ Any MSE owner can use it — no expertise needed

Scalable to all BIS product categories | Deployable as a free government portal for MSEs

63 million MSEs in India could benefit from instant, free, accurate standards discovery.

Team & Acknowledgements

Team: ByteCode

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Team Lead

Shraddha Ghadge

Member

Akshatha K S

Member

Tech Stack

Retrieval

FAISS + BM25 + RRF

Embeddings

all-MiniLM-L6-v2

Reranking

ms-marco-MiniLM-L-6-v2

LLM

Llama-3.1-8b via Groq

UI

Streamlit

Language

Python 3.9+

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